

A tree characterization one vertex above the Bougard–Joret counterexample boundary

Felipe Santibañez-Leal 

CAOS Research program

github.com/fsantibanezleal/CAOS_RESEARCH

2026-09-04

Version 0.01

PREPRINT · not peer reviewed · CAOS Research
Bougard–Joret research series, first paper · v0.01 · 2026-09-04
[Concept DOI: 10.5281/zenodo.22315251](https://doi.org/10.5281/zenodo.22315251) (latest version)
CC BY 4.0 · [Public sources and exact verification record](#)

Abstract

Let $f(n, \alpha, k)$ be the minimum size of a k -connected graph of order n and independence number α . Das and Gupta recently determined the boundary $n = \alpha + k$ and disproved the proposed Bougard–Joret formula there. We prove that $f(2k, k - 1, k) = k^2$ for every $k \geq 3$, and characterize all extremal graphs. They arise from a nonstar tree T on $k + 1$ vertices and an independent set of $k - 1$ vertices, partitioned into missing-neighbor fibers of sizes $d_T(t) - 1$. The proof is elementary and includes the exceptional connectivity case in which only two tree vertices survive a deletion. For $k \geq 4$ this settles the diagonal immediately above the known counterexample family; the numerical case $k = 3$ was already known. Exact finite controls use 86 nonstar trees, independent vertex-cut and max-flow checks, and a complete labeled order-six census. The universal statement follows from the proof, not from that census. The general interior and second regimes remain unresolved.

1 Problem, prior work and scope

All graphs are finite, simple and undirected. Write $e(G)$ for the number of edges, $\alpha(G)$ for the maximum size of an independent set, and $\kappa(G)$ for vertex connectivity. A graph is k -connected if it has more than k vertices and deletion of fewer than k vertices leaves it connected.

Bougard and Joret conjectured, under $n \geq 2\alpha$, $n \geq \alpha + k$, $\alpha \geq 2$, $k \geq 3$, that

$$f(n, \alpha, k) = \begin{cases} \lceil nk/2 \rceil, & n \leq k\alpha, \\ t(n, \alpha) + \lceil k\alpha/2 \rceil, & n > k\alpha, \end{cases}$$

where $t(n, \alpha)$ denotes the edge count of the balanced union of α cliques [1]. Das and Gupta determine the whole boundary $n = \alpha + k$; in particular, for $k \geq 4$ they prove

$$f(2k - 1, k - 1, k) = k^2 - 1, \tag{1}$$

with extremals exactly the joins of an independent $(k - 1)$ -set and a tree on k vertices [2]. The original disproof and this preceding-boundary characterization are their results.

The present work addresses the adjacent diagonal $(n, \alpha, k) = (2k, k - 1, k)$. It proves a formula and an equality characterization on this diagonal, not a corrected general conjecture. The numerical range $k \geq 4$ lies outside the identified cases in Section 6 of [1] and the boundary treated in [2]. For $k = 3$, the numerical value is already covered by $n = k\alpha$; it serves as a consistency case here. Targeted literature searches and both papers' final sections revealed no matching adjacent-diagonal characterization, but this is not an assertion of exhaustive bibliographic priority.

Our universal deductions are labeled [D], exact finite computations [MV], and suggested extensions [C]. The proof and verification are recorded in the repository's EXP-001 [3].

2 The tree construction and the extremal theorem

For a tree T on $k + 1$ vertices, let S be a disjoint independent set of size $k - 1$. Partition S into fibers $(S_t : t \in V(T))$ satisfying

$$|S_t| = d_T(t) - 1. \quad (2)$$

Keep all edges of T , and for each $s \in S_t$ add edges from s to every tree vertex except t . Call the resulting graph $G(T, (S_t))$. The required partition exists because

$$\sum_{t \in V(T)} (d_T(t) - 1) = 2k - (k + 1) = k - 1. \quad (3)$$

Theorem 1 (Adjacent exceptional strip, D). *For every integer $k \geq 3$,*

$$f(2k, k - 1, k) = k^2.$$

A graph is extremal for these parameters if and only if it is isomorphic to $G(T, (S_t))$ for a nonstar tree T on $k + 1$ vertices and fibers satisfying (2). Every extremal graph has this representation relative to each of its maximum independent sets.

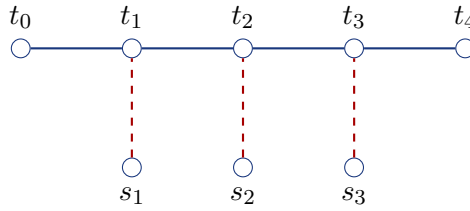


Figure 1: The $k = 4$ path construction. The solid path is $T = P_5$; the three dashed pairs are *nonedges*. Add every other pair between the two rows, and no pair within the lower row. This yields an eight-vertex, 16-edge extremal graph.

2.1 Regularity and independence

Every vertex of S has degree k . At a tree vertex t ,

$$d_G(t) = d_T(t) + (k - 1) - |S_t| = k.$$

Thus the graph is k -regular, has order $2k$, and has k^2 edges.

A nonstar tree of order $k + 1$ cannot contain an independent k -set: all its edges would then meet the remaining vertex, and connectedness would force a star. Consequently $\alpha(T) \leq k - 1$, and

also $d_T(t) \leq k - 1$ for every t . An independent set contained in either S or T therefore has size at most $k - 1$. An independent set meeting both parts contains only one tree vertex t , because each member of S has only one nonneighbor in T . Its other vertices lie in S_t , so its size is at most

$$1 + |S_t| = d_T(t) \leq k - 1.$$

Since S is itself independent, $\alpha(G) = k - 1$.

2.2 Connectivity

Let $X \subseteq V(G)$ satisfy $|X| < k$, and set

$$A = S \setminus X, \quad B = V(T) \setminus X.$$

There are at least two vertices in B . If A is empty, deletion of all $k - 1$ vertices of S uses the available deletion budget. Hence $G - X = T$ is connected.

Assume $A \neq \emptyset$ and $|B| \geq 3$. Any two vertices of A share a neighbor in B , since each misses at most one member of B . All of A , and all tree vertices adjacent to at least one member of A , therefore lie in a common component C . At most one tree vertex t can remain outside C : such a vertex must be the unique missed vertex of every member of A , so $A \subseteq S_t$. If t is disconnected from C , all its tree neighbors must have been deleted. All vertices of $S \setminus A$ were also deleted, and these sets are disjoint. Therefore

$$\begin{aligned} |X| &\geq k - 1 - |A| + d_T(t) \\ &\geq k - 1 - |S_t| + d_T(t) = k, \end{aligned}$$

a contradiction.

The case $|B| = 2$ requires a different argument. Write $B = \{u, v\}$. Exactly $k - 1$ tree vertices have been deleted, so every member of S survives. If a vertex of S misses a tree vertex other than u, v , it is adjacent to both u and v . Every other member of S is adjacent to at least one of them, proving connectedness. Otherwise the only nonempty fibers are S_u, S_v . By (2), every other tree vertex is a leaf. Since T is nonstar, u, v are exactly its two nonleaf vertices, and they must be adjacent: an internal vertex on their path would be a third nonleaf vertex. The surviving edge uv reconnects the graph. Thus G is k -connected in every case.

2.3 Optimality and all equality cases

Any k -connected graph on $2k$ vertices has minimum degree at least k and hence at least k^2 edges. The constructed graphs attain this lower bound, proving the numerical statement.

Conversely, let G be extremal, and let S be any maximum independent set, of order $k - 1$. Equality in the degree-sum lower bound forces k -regularity. The residual graph $H = G - S$ has $k + 1$ vertices and is connected, since fewer than k vertices were deleted. Each vertex of S misses exactly one vertex of H . There are $k(k - 1)$ edges between S and H , leaving

$$e(H) = k^2 - k(k - 1) = k.$$

Thus H is a tree. Writing S_t for the vertices that miss t , regularity gives

$$k = d_H(t) + (k - 1) - |S_t|,$$

so (2) holds. If H were a star, its k leaves would form an independent set in G , contrary to $\alpha(G) = k - 1$. This completes the proof of Theorem 1. $\square$

Remark 2. The theorem characterizes graphs relative to a marked maximum independent set. It does not assert a bijection between nonstar trees and unmarked graph isomorphism classes. Different maximum independent sets in the same graph need not give a uniquely determined representation without a further argument.

3 Boundary transition and a residual-graph viewpoint

Combining Theorem 1 with (1) gives the following precise transition.

Corollary 3 (D, using the preceding-boundary theorem of [2]). *For every $k \geq 4$,*

$$f(2k, k-1, k) - f(2k-1, k-1, k) = 1.$$

The excess over the degree-sum prediction drops from $\lfloor k/2 \rfloor - 1$ to zero.

Proof. The values are k^2 and $k^2 - 1$. At order $2k - 1$, the prediction is $\lceil k(2k-1)/2 \rceil = k^2 - \lfloor k/2 \rfloor$; at order $2k$ it is k^2 . $\square$

The mechanism differs from simply adding a vertex to a preceding-boundary extremal graph. On the original boundary every independent-set vertex is adjacent to every residual vertex. On the new diagonal it misses exactly one. The sizes of these missing-neighbor fibers compensate for the residual tree degrees, making every total degree equal to k . Connectivity must still be proved separately; the double-star case shows why a common-neighbor argument alone is insufficient.

4 Exact validation and adversarial controls

The computational record [3] is deliberately finite. It supports the proof by testing constructions and implementation failure modes; it is not the source of the universal quantifier. All arithmetic is integer and all choices are deterministic. The hypothesis was committed before the run.

Leaf augmentation generates every tree: deleting a leaf reduces any tree to a smaller tree. Canonical encodings obtained by minimizing rooted parenthesis encodings over all roots deduplicate isomorphic trees. All 92 trees of orders four through nine were generated; six stars failed the independence target, while all 86 nonstars passed. For every valid construction, exhaustive deletion of fewer than k vertices agreed with a separately implemented vertex-split max-flow test. Independence was verified both by include/exclude recursion and by direct subset enumeration. Deleting an edge from a valid construction was rejected by the connectivity checker.

An independent enumeration starts with *all labeled residual graphs* on $k + 1$ vertices with k edges and no isolates, rather than assuming a tree. Every allowed missing-neighbor assignment is enumerated from the degree sequence. The assignments for a fixed residual graph differ by explicit permutations of S , so expensive invariants are checked once per residual graph. Table 1 records the results; the accepted graphs are exactly the connected nonstars.

A separate exhaustive census visits all $2^{15} = 32768$ labeled graphs on six vertices. It finds 60 extremal graphs for $(6, 2, 3)$ and verifies the tree characterization relative to every maximum independent set in each. The result is consistent with the triangular-prism base case. The experiment emits deterministic JSON checkpoints and a source hash; its regression test reruns the certificate into a temporary directory and compares the complete result.

An independent adversarial reasoning review caught the initially omitted $|B| = 2$ case before computation. The committed hypothesis and the proof above include its repair. The same review rejected an unsupported uniqueness interpretation for unmarked tree representations. These checks are not peer review or a proof-assistant formalization.

k	Residuals	Assignments	Nonstar trees	Stars	Disconnected
3	16	28	12	4	0
4	135	605	120	5	10
5	1581	22986	1290	6	285

Table 1: Exact labeled residual enumeration [MV]. Assignment totals include rejected residuals; they are not counts of unmarked extremal graphs.

5 Remaining questions and reproducibility

The entire shell $n = \alpha + k + 1$ remains outside this theorem when $k - \alpha \neq 1$. In degree-sum equality cases, a maximum independent set again yields missing-neighbor fibers, but the residual graph need not be a tree [C, proposed research direction]. A new argument must control its connectivity and independence number. Neither the corrected general first regime nor the second regime is established here.

The experiment directory contains `hypothesis.md`, `proof.md`, `run.py`, `verdict.md`, and `artifacts/certificate.json`. From the repository root, run:

```
python problems/combinatorics/bougard-joret/
experiments/EXP-001-tree-strip/run.py
```

The displayed path is wrapped typographically; the shell command uses one uninterrupted path. The checker uses only the Python standard library. A portfolio-wide literature review and its limitations are stored separately in the problem context; they are not claims of this theorem. The experiment’s verdict, rather than a dashboard or manuscript summary, owns the computational evidence.

References

- [1] N. Bougard and G. Joret. Turan’s theorem and k -connected graphs. *Journal of Graph Theory* 58 (2008), 1–13. doi:10.1002/jgt.20289. Author version: <https://gjoret.be/papers/turan.pdf>.
- [2] J. Das and S. Gupta. Counterexample to the Bougard–Joret conjecture. arXiv:2608.18828v1, 19 August 2026. <https://arxiv.org/abs/2608.18828v1>.
- [3] F. Santibañez-Leal. CAOS Research, Bougard–Joret EXP-001: tree-strip proof and exact certificate, 4 September 2026. [Public experiment record](#). Preprint version DOI: [10.5281/zenodo.22315252](https://doi.org/10.5281/zenodo.22315252).